SHUD

2022-12-24

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2. Advisor Mentor
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5.
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7.

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Chapter 1

1.1

1.1.1

1.
2.
3.
4. “ - - ”
5.
6.
7.

1.1.2

(www.nieer.cas.cn) 2 6 1 2

1.2

1.3

1.3.1

1.3.2

1.3.3

1.3.4

1.3.5

1.3.6 /

1.3.7

1.4

1.4.1

“ ” SHUD https://www.shud.xyz 2017 (Pennsylvania State University) 2017-2020 (UC Davis) 2020

Chapter 2

2.1 (Adacemic Advisor)

1.
2.

2.2 (Mentor)

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2.3

2.3.1

1.1.1

1. 2.

“ ”

1.
2. A B / A B
3.
4.

2.3.2

-12		
-6		
0-6		
7-12	1	
13-18	1.5	
19-24	2	
25-30	2.5	
31-36		:-)

2.3.3

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  - /
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  -
  - DEM D8
-
  -
  -
  -
  -
-
  - / Duipuit Assumption
  - van Genuchten
  - /
-

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  - /
  -
  - CFL
  -

-
  -
  - /
  -

- **R/Python**
  - ( )
  - R/Python (shapefile, GeoTiff HDF, NetCDF)
  - (/)
  - ggplot, rgl

2.3.4

- Beven, K., 2012. Rainfall-Runoff Modelling: The Primer: Second Edition, John Wiley & Sons, Ltd, Chichester, UK. https://doi.org/10.1002/9781119951001
- Sorooshian, S., Hsu, K.-L., Coppola, E., Tomassetti, B., Verdecchia, M., Visconti, G. (Eds.), 2008. Hydrological Modelling and the Water Cycle, Water Science and Technology Library. Springer Berlin Heidelberg, Berlin, Heidelberg. https://doi.org/10.1007/978-3-540-77843-1
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- , 2009. . , .
- , , , 2005. . , .
- Kang-tsung Chang . 2019. . . .
- McDonnell, J.J., 2019. Navigating an Academic Career, Special Publications. Wiley. https://doi.org/10.1002/9781119642206

Chapter 3

3.1

- (2022) 29600 49000
- PI

3.2

PI

Chapter 4

4.1

320

4.2

4.2.1

100TB

Synology Drive NAS sftp, Synology Drive rsync

4.2.2 /

-
  - PI PI
  - NAS
  - NAS NAS
-
  -
  - NAS

4.2.3 NAS

NAS

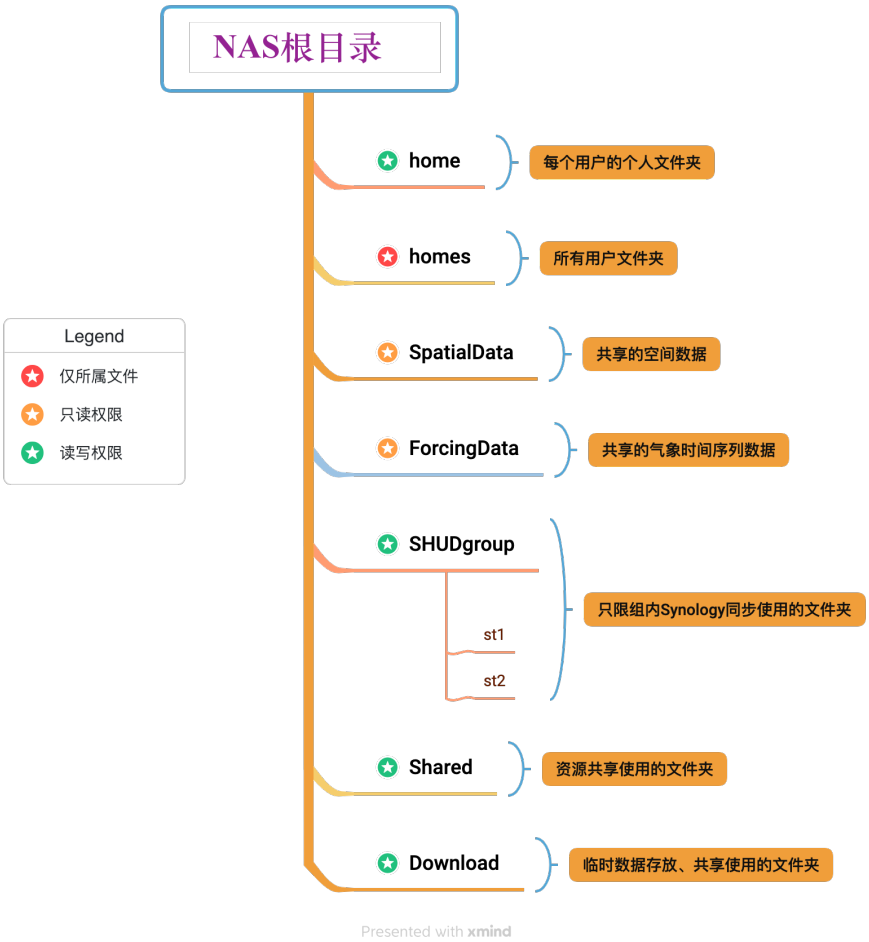

Figure 4.1: NAS

---

homes	
home	10TB
SpatialData	10TB
ForcingData	10TB

SHUDgroup	10TB
Shared	10TB
Download	

4.2.4 NAS

- NAS_IP_Address: NAS IP PI
- username[ ]
- : [ ]
- NAS https://shudxyz.direct.quickconnect.to:5318

4.2.5 NAS

- Mac OS
  - Finder CMD+k
  - `afp://[NAS_IP_Address]` Connect
  -
  -
- Windows

4.2.6 Synology Drive

Synology Drive NAS / “ ” “ ” “ ”

4.3

4.4

4.5

4.6

- http://www.ncdc.ac.cn

- http://www.tpdc.ac.cn
- https://www.resdc.cn
- NASA Earth Observation Data https://www.earthdata.nasa.gov
- USGS EarthExplorer https://earthexplorer.usgs.gov
- World Climate Research Programme, CMIP6 data https://esgf-node.llnl.gov/projects/cmip6/
- World Soil Information https://www.isric.org
- USDA Web Soil Survey https://websoilsurvey.sc.egov.usda.gov/App/HomePage.htm
- FAO Soil Portal https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/en/
- MERIT Hydro: global hydrography datasets http://hydro.iis.u-tokyo.ac.jp/~yamadai/MERIT_Hydro/
- MERIT-Basins https://www.reachhydro.org/home/params/merit-basins
- National Hydrography Dataset (NHD) https://www.usgs.gov/national-hydrography/national-hydrography-dataset

4.7

4.7.1 Windows Server

CPU	SA5212 H5 Xeon 6133 16GB x12 U.2 NVME	12 2U CPU 20 40 2.5GHz DDR4 2666Mhz 2TB 10Gbps	16 GBx12	CPU 6 12

4.7.1.1

- Windows Server 2016 GUI
- Office: Microsoft Office 365
- Notepad++, vim.
- GIS ArcGIS, QGIS
- Visual Studio Code
- R: R, Rstudio
- Octave

4.7.1.2

- userx

- zhangsan

4.7.2 Ubuntu

CPU	SA5212 H5 Xeon 6133 16GB x12 U.2 NVME	12 2U CPU 20 40 2.5GHz DDR4 2666Mhz 16 GBx12 2TB 10Gbps	CPU 6 12

4.7.2.1

- Ubuntu 20.04 GUI
- GIS QGIS
- R: R, Rstudio

4.7.2.2

- userx: sudo
- zhangsan

4.7.3

		8 1 1 /	56G IB ,10G
	SA5212 H5 2U	2x 6133 192GB 2T U.2 NVME	
	2288 V6	2x 6318Y 128 2TB U.2 NVME	
	2288 V6	12X16TB, RAID 6	140TB
IB	SX6025 IBM	36x56Gbps 48x10Gbps 4x40Gbps	InfiniBand

4.7.3.1

4.7.3.2

4.7.3.3 Slurm

Chapter 5

5.1

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5.2

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5.3 /

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Chapter 6

6.1

1. —— “ ”
2. ——
3. ——
4. —— Academic Credit
5. ——
6. ——
7. –
8. ——

6.2

1. ——
2. ——
3. ——
4. ——

6.3

1. ——
2. ——
3. ——

Chapter 7

7.1

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7.1.1

https://www.scribbr.com/research-process/research-question-examples/

7.2

7.2.1

7.2.1.1

- GDP

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7.2.1.2

- DEM
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- /
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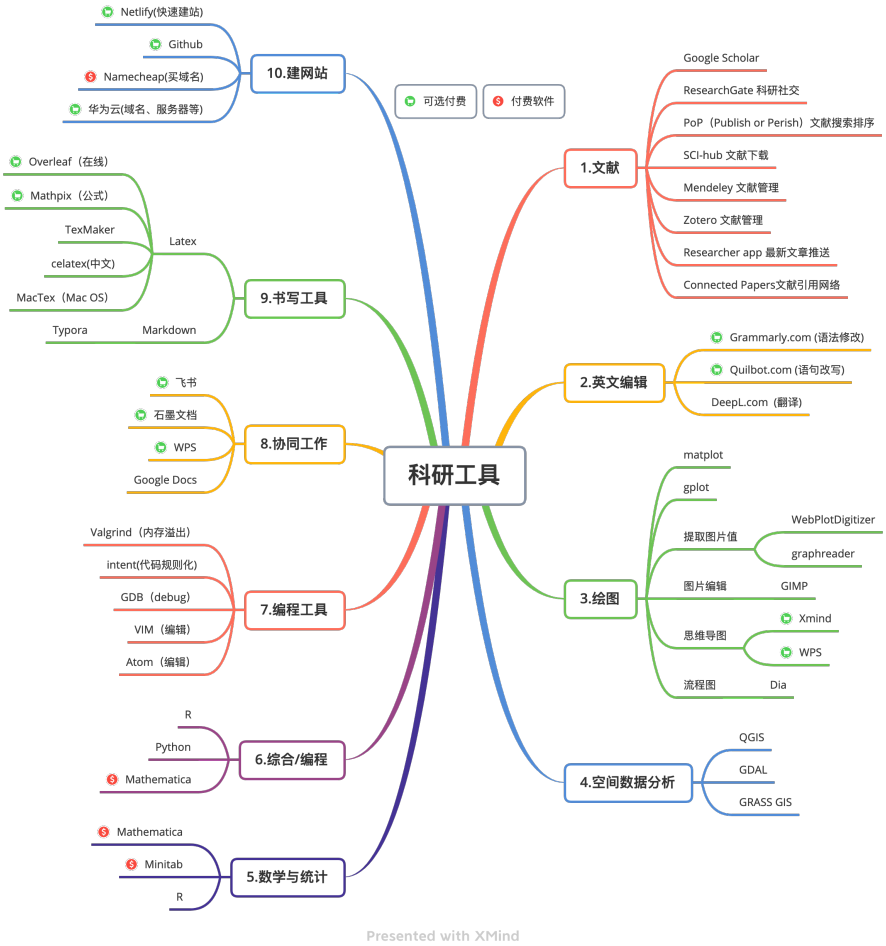
7.2.1.3

- GLDAS
- CMFD
- PRISM

Chapter 8

8.1


科研工具
可选付费
付费软件
1.文献
Google Scholar
ResearchGate 科研社交
PoP（Publish or Perish）文献搜索排序
SCI-hub 文献下载
Mendeley 文献管理
Zotero 文献管理
Researcher app 最新文章推送
Connected Papers文献引用网络
2.英文编辑
Grammarly.com (语法修改)
Quilbot.com (语句改写)
DeepL.com（翻译）
3.绘图
matplot
gplot
提取图片值
WebPlotDigitizer
graphreader
图片编辑
GIMP
思维导图
Xmind
WPS
流程图
Dia
4.空间数据分析
QGIS
GDAL
GRASS GIS
5.数学与统计
Mathematica
Minitab
R
6.综合/编程
R
Python
Mathematica
7.编程工具
Valgrind（内存溢出）
intent(代码规则化)
GDB（debug）
VIM（编辑）
Atom（编辑）
8.协同工作
飞书
石墨文档
WPS
Google Docs
9.书写工具
Latex
Overleaf（在线）
Mathpix（公式）
TexMaker
celatex(中文)
MacTex（Mac OS）
Markdown
Typora
10.建网站
Netlify(快速建站)
Github
Namecheap(买域名)
华为云(域名、服务器等)
Presented with XMind

8.1.1 QGIS

QGIS

https://www.qgis.org/en/site/forusers/download.html

8.1.2 XMind

https://xmind.cn/download/

8.1.3 Mendeley Desktop

https://www.mendeley.com/download-reference-manager

8.2

8.2.1 Windows RDP

8.2.2 VNC Ubuntu

8.2.3 VNC Mac

8.3 ssh

ssh

```
ssh [username@remotehost]
```

, y

- **-p 22** - 22
- **-Y** - GUI
- **-i** [**file**] -

8.3.1

ssh-keygen SSH

```
ssh-keygen
```

```
ls ~/.ssh/
cat ~/.ssh/id_rsa
```

,

```
ssh-copy-id [username@remotehost]
```

SSH

```
ssh -i ~/.ssh/id_rsa [username@remotehost]
```

8.4 Linux

8.4.1

userx

```
sudo useradd -s /bin/bash -d /home/userx/ -m -G sudo userx
```

userx

```
su - userx
```

8.5 Slurm

8.6 R

R rSHUD(https://github.com/shud-System/rshud), AutoSHUD(https://github.com/shud-System/autoshud) (https://www.shuddata.com) R

R R https://www.shud.xyz/bookr/

8.7 Python

Chapter 9

		2021.09	;
		2022.10	;
	( )	2022.10	;

( )	2021.10	2022.06
( )	2021.01	2021.03